# Module 2 Assignment 3

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## Stratified Random Sampling

### Assignment Details

#### Purpose

The goal of this assignment is to understand, apply, and interpret a stratified random sampling method and calculations.

#### Task

Write R code to successfully answer each question below or write text to successfully answer the question.

#### Criteria for Success

- Code is within the provided code chunks
- Code chunks run without errors
- Code produces the correct result
  - Code attempts will get half points
  - Code that produces the correct answer will receive full points
- Text answers correctly address the question asked

#### Due Date

March 31 before lab

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### Assignment Questions

We have been asked to go out and sample for grasshoppers. The location we are sampling has 3 different vegetation types: riparian, shrubland, and grassland. We have decided to use a stratified random sample design for our survey.

1. In 1-2 sentences, explain why we have chosen stratified random sampling as opposed to simple or systematic in this scenario? What benefit does it provide us?

   *Answer:* taking into account that different vegetation types might have different means, totals, etc. trying to minimize heterogeneity, makes our estimates more precise

### Part 1: Allocation of Sample Units

We have a sampling frame with 1000 sample units. We decide sample 100 of those units. We have our 3 strata which we have determined based on the vegetation type (we will called them A, B, and C for simplicity for the rest of the assignment).

How do we decide how many sampling units to sample from each stratum?

2. Create an object, `N`, which represents the total number of sample units in our *sampling frame*. We also need to create `n`, which is the total number of sample units in our *sample*.

```
N <- 1000
n <- 100
```

We have determined that Stratum A has 200 sampling units, Stratum B has 500 sampling units, and Stratum C has 300 sampling units.

3. Add the correct values to the lines of code below to create objects for future use.

```
N_A <- 200
N_B <- 500
N_C <- 300
```

Due to a pilot study done last year, we have an estimate of the amount of variation (standard dev.) for each stratum.

```
sd_A <- 9
sd_B <- 4
sd_C <- 7
```

We also have an estimate of the time-cost (hours) it takes to survey each unit.

```
cost_A <- 1.5
cost_B <- 2
cost_C <- 4
```

### Optimal Allocation

We want to use the optimal allocation equation to calculate how many sample units each stratum should have. Let's do this incrementally.

4. Start by calculating the numerators for each stratum. Each numerator looks like this:

$$\frac{N_i * s_i}{\sqrt{c_i}}$$

   Remember to use objects you've created rather than typing out specific numbers into the equation.

```
(opt_allo_A <- N_A * sd_A / sqrt(cost_A))
```

```
## [1] 1469.694
```

```
(opt_allo_B <- N_B * sd_B / sqrt(cost_B))
```

```
## [1] 1414.214
```

```
(opt_allo_C <- N_C * sd_C / sqrt(cost_C))
```

```
## [1] 1050
```

5. Calculate the denominator of the equation by adding all of the values above together.

```
opt_allo_total <- opt_allo_A + opt_allo_B + opt_allo_C
opt_allo_total
```

```
## [1] 3933.907
```

6. Calculate how many sampling units we should optimally sample in each stratum. Use the `round()` function to round to the nearest whole number.

Make sure to *save the rounded value* to use for the remainder of the questions.

```
# if students don't save the ROUNDED output, their answers below end up ever-so-slightly off
n_A <- n * (opt_allo_A / opt_allo_total) # 37
(n_A <- round(n_A))
```

```
## [1] 37
```

```
# OR #
(n_B <- round(n * (opt_allo_B / opt_allo_total))) # 36
```

```
## [1] 36
```

```
(n_C <- round(n * (opt_allo_C / opt_allo_total))) # 27
```

```
## [1] 27
```

7. If we did not have the pilot study telling us about the variation of the populations in each strata, we could have allocated very differently:

   - stratum A = 24
   - stratum B = 53
   - stratum C = 23

   Why is this? What would this allocation have meant for the precision of our estimates?

   *Answer:* without taking standard deviation into account, we would have had far less precise estimates because we would have allocated more samples to strata with low variability; cost is also factored in

## Part II: Stratified Random Sampling

### Data

Hooray! We have gone out and sampled the number of grasshoppers per sample unit, allocating our sample units optimally (including variation in population).

Now we want to calculate our population parameters:

- mean grasshoppers per sampling unit (`y_bar`)
- total grasshoppers in the population (`tau_hat`)

Let's load our "grasshoppers.csv" data into our workspace using the `read.csv()` function and save the data frame as an object called `grasshoppers`.

```
grasshoppers <- read.csv("../data_raw/grasshoppers.csv")
```

We can explore the data frame to get an idea of what each row and column represent.

```
str(grasshoppers)
```

```
## 'data.frame':    100 obs. of  3 variables:
##  $ X      : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ stratum: chr  "A" "A" "A" "A" ...
##  $ abund  : int  34 26 32 29 35 38 38 28 31 24 ...
```

Based on what I've taught you in R thus far, the easiest way to do our calculations will be to have three different data frames, one for each stratum.

```
stratumA <- grasshoppers[grasshoppers$stratum == "A",]
stratumB <- grasshoppers[grasshoppers$stratum == "B",]
stratumC <- grasshoppers[grasshoppers$stratum == "C",]
```

Run the following code chunk. What do these numbers represent?

*(Hint: do these numbers match your values from question 6? If not, you probably want to revisit your calculations up in the previous section!)*

```
nrow(stratumA)
```

```
## [1] 37
```

```
nrow(stratumB)
```

```
## [1] 36
```

```
nrow(stratumC)
```

```
## [1] 27
```

The data in `stratumA`, `stratumB`, and `stratumC` are the data collected when we surveyed each sample unit as allocated in Question 6.

Below, where you need to reference $N$, $N_i$, $n$, or $n_i$ in an equation, you can use the values you calculated in Questions 1-6 above.

## Calculating Grasshopper Abundance and Density

NOTE: Open up your previous assignments; they are likely about to come in handy! I would also recommend opening up the lecture slides. Remember to download the slide decks to see all of the equations or use the PDF versions.

8. Before we begin calculating, stop for a moment to reflect and predict. For stratified sampling, we have a lot of steps to take before we get to our ultimate goal: estimates of density and abundance of the entire grasshopper population, along with the amount of uncertainty in those estimates.

   Take a minute to skim through the lecture slides and remind yourself of the process of how we go about calculating the estimates of population parameters and the amount of uncertainty in those estimates. Describe (either in sentences or in bullet points) what the best way to proceed through our calculations will be.

   *Note: This question in graded for completion only, not for accuracy.*

   1. calculate population parameter estimates for each stratum
   2. calculate measures of uncertainty (variance of estimates of mean, total) for each stratum
   3. calculate overall population total and overall variance of the estimate of the total
   4. calculate overall population mean and overall variance of the estimate of the mean

## Calculations by Stratum

### Step 1: Calculating Sample Statistics (per stratum)

For our first step, we will calculate for each stratum the mean number of grasshoppers, the total number of grasshoppers, and the sample variances.

9. Use the code from Module 2 Assignment 2 to calculate the following for each stratum:

   a. mean grasshoppers per stratum (1 point)

```
# take the mean of the `abund` column
(mean_A <- mean(stratumA$abund))
```

```
## [1] 31.86486
```

```
(mean_B <- mean(stratumB$abund))
```

```
## [1] 35.47222
```

```r
(mean_C <- mean(stratumC$abund))
```

```
## [1] 40
```

b. total grasshoppers per stratum (`tau_hat`) (1 point)

```r
(tau_hat_A <- N_A * mean_A)
```

```
## [1] 6372.973
```

```r
(tau_hat_B <- N_B * mean_B)
```

```
## [1] 17736.11
```

```r
(tau_hat_C <- N_C * mean_C)
```

```
## [1] 12000
```

c. sample variance per stratum (use the `var` function) (1 point)

```r
# similar to (a) but calculating the variance instead of the mean
(var_A <- var(stratumA$abund))
```

```
## [1] 27.17568
```

```r
(var_B <- var(stratumB$abund))
```

```
## [1] 36.82778
```

```r
(var_C <- var(stratumC$abund))
```

```
## [1] 31.69231
```

**Step 2: Calculating Uncertainty (per stratum)**

Our next step is to calculate the uncertainty in our estimates for each stratum.

10. To do this, calculate the following:

    a. variance of the estimate of the mean per stratum (var_y_bar). Remember that we need to use the finite population correction factor and to use only objects already created, not plain numbers! In this specific instance, please use `var_A` instead of `sd_A^2` (same for B and C). (2 points)

```r
# did they include the population correction factor?
# parathenses get a little tricky here
# students also forget that they need the * sometimes
# also need to use var_X rather than sd_X^2 because the sd_X values are made up
(var_y_bar_A <- ((N_A-n_A)/N_A) * (var_A / n_A))
```

```
## [1] 0.5985993
```

```r
(var_y_bar_B <- ((N_B-n_B)/N_B) * (var_B / n_B))
```

```
## [1] 0.9493383
```

```r
(var_y_bar_C <- ((N_C-n_C)/N_C) * (var_C / n_C))
```

```
## [1] 1.068148
```

    b. variance of the estimate of the total abundance (var_tau_hat). You might need to go fishing for this equation in one of the earlier powerpoints. (1 point)

```r
(var_tau_hat_A <- N_A^2 * var_y_bar_A)
```

```
## [1] 23943.97
```

```r
(var_tau_hat_B <- N_B^2 * var_y_bar_B)
```

```
## [1] 237334.6
```

```r
(var_tau_hat_C <- N_C^2 * var_y_bar_C)
```

```
## [1] 96133.33
```

**Step 3: Calculating Total Abundance Statistics**

Congratulations! The hardest parts of the calculation are over (thank goodness, right?).

Next, we will calculate the total abundance and the variance of the total abundance.

11. Calculate the following:

    a. total abundance (1 point)

```r
tau_hat <- tau_hat_A + tau_hat_B + tau_hat_C
tau_hat
```

```
## [1] 36109.08
```

    b. variance of the total abundance estimate (1 point)

```r
var_tau_hat <- var_tau_hat_A + var_tau_hat_B + var_tau_hat_C
var_tau_hat
```

```
## [1] 357411.9
```

**Step 4: Calculating Population Means, etc.** Finally, we can now calculate the overall population mean and variance of the estimate of the overall population.

12. Calculate the following:

    a. overall population mean (1 point)

```r
y_bar <- tau_hat / N
y_bar
```

```
## [1] 36.10908
```

    b. variance of the estimate of the overall population mean (1 point)

```r
var_y_bar <- var_tau_hat / N^2
var_y_bar
```

```
## [1] 0.3574119
```

**Conclusion** Ok, that was a lot of calculating! Let's summarize and make sure we know what just happened. We were ultimately able to calculate estimates for the parameters we were interested in: the population mean and the population total along with the amount of uncertainty in both of those estimates.

12. Based on our calculations above, answer the following questions with the name of the R object and its corresponding value for each calculation. (2 points)

    a. the number of grasshoppers per sample unit: `y_bar`, 36.12

    b. total number of grasshoppers in the population: `tau_hat`, 36109

c. measure of uncertainty in the # of grasshoppers per sample unit: `var_y_bar`, 0.357

d. measure of uncertainty in the total # of grasshoppers: `var_tau_hat`, 357411.9